

D401

Introduction to Epidemiology

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A town of 10,000 residents had 400 people with diagnosed diabetes on January 1. During the year, 144 new cases were diagnosed. The town's hospital added a new endocrinology wing that year. What was the annual incidence per 1,000 people at risk?

- A. 54.4 per 1,000
- B. 14.4 per 1,000
- C. 15 per 1,000
- D. 40 per 1,000

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A new treatment cures a chronic disease much faster, but the number of new cases each year does not change. What happens to prevalence?

- A. It stays the same, because incidence did not change.
- B. Incidence falls instead.
- C. It falls, because people have the disease for less time.
- D. It rises, because more people are treated.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

Researchers in a coastal state identify 200 people with a rare liver cancer and 400 similar people without it, then review their past work records for chemical exposure. The study was funded by a large foundation. What design is this?

- A. Cross-sectional
- B. Case-control
- C. Prospective cohort
- D. Randomized controlled trial

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

In a 10-year study, 30 of 1,000 people who smoked and 20 of 2,000 people who did not smoke developed a lung disease. The study was run at five universities. What is the relative risk for smokers?

- A. 3.0
- B. 1.5
- C. 0.33
- D. 2.0

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

The risk of a disease is 2% in unexposed people, and the relative risk for the exposed is 4. What is the risk in the exposed group?

- A. 8%
- B. 4%
- C. 0.5%
- D. 6%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

After a rainy summer, a county sees a rise in West Nile virus. The county's public health director is a former Army physician. Health workers drain standing water in tires and gutters. Which link in the chain of infection does this target?

- A. The agent, by killing the virus directly.
- B. Only killing the virus can stop the outbreak.
- C. The susceptible host, by building immunity.
- D. Mode of transmission (the mosquito vector) and its environment

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A new screening test is tried on 1,000 adults at a community health fair. Of 200 people with the disease, 180 test positive. Of 800 people without it, 760 test negative. The fair also offered free blood pressure checks. What is the test's specificity?

- A. 5%
- B. 95%
- C. 82%
- D. 90%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

A clinic wants a screening test whose negative result reliably rules out a dangerous disease. Which property matters most?

- A. High prevalence
- B. High specificity
- C. Low cost
- D. High sensitivity

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A state usually sees about 50 cases of a certain foodborne illness each summer. This July it has 300 cases, all within the state, many traced to a popular fair with a new ride. How should this be described?

- A. An epidemic (outbreak)
- B. Pandemic
- C. Endemic
- D. Not unusual, because the illness occurs every summer.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A clinic serving 5,000 patients offers colonoscopy to adults over 45 who have no symptoms. The clinic just renovated its lobby. What level of prevention is this?

- A.** Tertiary prevention
- B.** Primary, because patients have no symptoms.
- C.** Secondary prevention
- D.** Primary prevention

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

Which activity is tertiary prevention?

- A.** A school anti-smoking program.
- B.** A flu vaccine clinic.
- C.** Mammogram screening.
- D.** A cardiac rehab program for people after a heart attack.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A city notices that ice cream sales and drowning deaths both rise in the same months. A local reporter who won a journalism award writes that ice cream causes drowning. What best explains the link?

- A.** Any two things that rise together are always unrelated.
- B.** Hot weather is a confounder that increases both ice cream sales and swimming.
- C.** The link must be true because it was reported by an award winner.
- D.** Ice cream causes cramps that lead to drowning.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A town of 10,000 residents had 400 people with diagnosed diabetes on January 1. During the year, 144 new cases were diagnosed. ~~The town's hospital added a new endocrinology wing that year.~~ What was the annual incidence per 1,000 people at risk?

Target word: incidence rate **Decides:** Only people without diabetes (9,600) are at risk.

A	Confusion set	$(400 + 144) / 10,000$ is end-of-year prevalence, not incidence.
B	Dropped element	$144 / 10,000$ keeps existing cases in the at-risk group.
C	Correct	$144 / 9,600 = 0.015 = 15$ per 1,000.
D	Wrong stem number	The starting prevalence, $400 / 10,000$.

2. Answer C Concept 1 • transfer

A new treatment cures a chronic disease much faster, but the number of new cases each year does not change. What happens to prevalence?

Target word: reverse: prevalence from duration **Decides:** $\text{Prevalence} \approx \text{incidence} \times \text{duration}$.

A	Dropped element	Leaves out duration.
B	Confusion set	Curing people does not change how many new cases occur.
C	Correct	Shorter duration lowers prevalence at the same incidence.
D	Inverted operation	Treatment that cures shortens duration and lowers prevalence.

3. Answer B Concept 2

Researchers in a coastal state identify 200 people with a rare liver cancer and 400 similar people without it, then review their past work records for chemical exposure. ~~The study was funded by a large foundation.~~ What design is this?

Target word: design from sampling **Decides:** Selected by outcome, then looked back at exposure.

A	Confusion set	Measures exposure and outcome at a single time without selecting by outcome.
B	Correct	Starts with the outcome and looks back at exposure.
C	Inverted operation	A cohort selects by exposure and follows forward.
D	Adjacent category	No exposure was assigned.

4. Answer A Concept 3

In a 10-year study, 30 of 1,000 people who smoked and 20 of 2,000 people who did not smoke developed a lung disease. ~~The study was run at five universities.~~ What is the relative risk for smokers?

Target word: compute RR **Decides:** Risks: $30/1,000 = 3\%$; $20/2,000 = 1\%$.

A	Correct	$0.03 / 0.01 = 3.0$.
B	Dropped element	$30 / 20$ divides counts and drops the group sizes.
C	Inverted operation	Divides unexposed by exposed.
D	Confusion set	$3\% - 1\% = 2$ percentage points is the risk difference, not RR.

5. Answer A Concept 3 • transfer

The risk of a disease is 2% in unexposed people, and the relative risk for the exposed is 4. What is the risk in the exposed group?

Target word: solve for exposed risk **Decides:** Exposed risk = RR × unexposed risk.

A	Correct	$4 \times 2\% = 8\%$.
B	Wrong stem number	The RR read as a percentage.
C	Inverted operation	Divides the baseline by the RR.
D	Inverted operation	Adds the RR to the baseline risk.

6. Answer D Concept 4

After a rainy summer, a county sees a rise in West Nile virus. ~~The county's public health director is a former Army physician.~~ Health workers drain standing water in tires and gutters. Which link in the chain of infection does this target?

Target word: which link is broken **Decides:** Draining water removes mosquito breeding sites that carry the virus.

A	Adjacent category	Draining water does not kill the virus itself.
B	Absolute language	Breaking any link can control spread.
C	Confusion set	Vaccination targets the host; draining water does not.
D	Correct	Fewer mosquitoes means fewer bites that carry the virus.

7. Answer B Concept 5

A new screening test is tried on 1,000 adults at a community health fair. Of 200 people with the disease, 180 test positive. Of 800 people without it, 760 test negative. ~~The fair also offered free blood pressure checks.~~ What is the test's specificity?

Target word: compute specificity **Decides:** True negatives / all without disease.

A	Inverted operation	$40 / 800$ is the false positive rate.
B	Correct	$760 / 800 = 95\%$.
C	Adjacent category	$180 / 220$ is the positive predictive value.
D	Confusion set	$180 / 200$ is sensitivity.

8. Answer D Concept 5 • transfer

A clinic wants a screening test whose negative result reliably rules out a dangerous disease. Which property matters most?

Target word: reverse: from purpose to property **Decides:** Ruling out needs few false negatives.

A	Confusion set	A population feature, not a test property.
B	Inverted operation	Specificity helps rule in, not out.
C	Adjacent category	Important, but not the property asked.
D	Correct	A highly sensitive test rarely misses disease, so a negative rules it out.

9. Answer A Concept 6

A state usually sees about 50 cases of a certain foodborne illness each summer. This July it has 300 cases, all within the state, ~~many traced to a popular fair with a new ride~~. How should this be described?

Target word: classify the pattern **Decides:** Far more than expected, in one area.

A	Correct	Cases far exceed the usual baseline in one area.
B	Adjacent category	Pandemics span countries or continents.
C	Inverted operation	Endemic means the usual expected level, which is 50.
D	Dropped element	Ignores the jump from 50 to 300.

10. Answer C Concept 7

A clinic ~~serving 5,000 patients~~ offers colonoscopy to adults over 45 who have no symptoms. ~~The clinic just renovated its lobby~~. What level of prevention is this?

Target word: prevention level **Decides:** Screening finds existing disease early.

A	Inverted operation	Tertiary manages known disease.
B	Absolute language	No symptoms does not mean no disease; screening is secondary.
C	Correct	Screening detects disease early, before symptoms.
D	Confusion set	Primary prevents disease from starting.

11. Answer D Concept 7 • transfer

Which activity is tertiary prevention?

Target word: reverse: pick the tertiary example **Decides:** Limiting harm from existing disease.

A	Inverted operation	Primary prevention.
B	Confusion set	Primary prevention.
C	Adjacent category	Secondary prevention.
D	Correct	Reduces disability from existing disease.

12. Answer B Concept 8

A city notices that ice cream sales and drowning deaths both rise in the same months. A local reporter ~~who won a journalism award~~ writes that ice cream causes drowning. What best explains the link?

Target word: confounding **Decides:** A third factor drives both.

A	Absolute language	Associations can be causal; they must be tested.
B	Correct	The association is not causal.
C	Camouflage	The reporter's award is not evidence.
D	Inverted operation	Accepts the causal claim.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	4	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	11	
Absolute language	Always, never, guarantee — where judgment applies.	3	
Confusion set	Every option is real; the test is exact role-matching.	9	
Camouflage	Answers a stem fact that was never the question.	1	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Incidence vs. prevalence	Incidence counts new cases among people at risk over a period; prevalence counts all existing cases at a point in time; prevalence $\approx$ incidence $\times$ average duration.	Counting existing cases as new, or keeping people who already have the disease in the at-risk denominator.	A starting count of cases plus new cases during a period.
2	Cohort vs. case-control design	A cohort study starts with exposure and follows people forward to see who gets the outcome (gives relative risk); a case-control study starts with people who have and do not have the outcome and looks back at exposure (gives an odds ratio).	Naming a study by its time direction only, or getting a relative risk from a case-control study.	Whether the study selects people by exposure or by outcome.
3	Relative risk	Relative risk = risk in the exposed $\div$ risk in the unexposed, where each risk is cases divided by the group's size; RR above 1 means higher risk, below 1 means lower.	Dividing raw case counts, or subtracting the risks and calling it relative.	Two groups with different sizes and case counts.
4	Epidemiologic triangle and chain of infection	Disease results from agent, host, and environment interacting; control breaks the weakest link in the chain — reservoir, portal of exit, mode of transmission, portal of entry, or susceptible host.	Believing only killing the agent can stop spread.	A control measure to classify, such as hand hygiene, vaccination, or draining standing water.
5	Sensitivity vs. specificity	Sensitivity = true positives $\div$ all who have the disease; specificity = true negatives $\div$ all who do not; a highly sensitive negative result helps rule a disease out.	Mixing them up, or using test positives in the denominator (that is predictive value).	A 2 $\times$ 2 table or a question about ruling in or out.
6	Endemic, epidemic, pandemic	Endemic is the usual, expected level in an area; an epidemic is more cases than expected in an area; a pandemic is an epidemic spread across countries or continents.	Using "pandemic" for any serious outbreak or "endemic" for a harmless disease.	A comparison to the usual baseline and a geographic range.
7	Levels of prevention	Primary prevention stops disease before it starts (vaccines, smoking prevention); secondary finds it early (screening); tertiary limits harm from existing disease (rehab, disease management).	Calling screening primary prevention because it happens before symptoms.	Whether the person already has the disease and whether it is known.

#	Concept	Rule	Trap	Tell
8	Association is not causation	A third factor linked to both exposure and outcome — a confounder — can create an association that is not causal; control it through design or analysis.	Concluding that one factor causes another because they rise together.	Two things that rise together in the same season or group.

Course scope (public description): Introduction to Epidemiology provides an overview of the determinants of communicable, viral, and chronic diseases. Students also will study various other conditions and the impact to public health. Using problem-based inquiry, students will analyze real- world public health problems by examining the distribution and patterns of data, selecting the methods to gather evidence, interpreting the information, and analyzing the trends to support decision making. There are no prerequisites to this course, but students are highly encouraged to adhere to the standard path, whose content is scaffolded to enhance the learning experience of this course.